

# Lesson 7: Repeated Measurement and Probability

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Reinforcing superposition, sampling, and histogram interpretation

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# Learning Outcomes

- Repeat the one-qubit superposition experiment.
- Relate measurement counts to probabilities.
- Understand why visualization is useful for beginners.

## Lesson 7 Position in the Sequence

- The notebook repeats the Hadamard measurement experiment using 1000 shots.
- The focus is interpretation of the output and histogram.
- This repetition is deliberate: fluency with measurement is essential before multi-qubit concepts.

# Core Qiskit Code

```
qc = QuantumCircuit(1, 1)
qc.h(0)
qc.measure(0, 0)
simulator = Aer.get_backend('aer_simulator')
job = simulator.run(qc, shots=1000)
result = job.result()
counts = result.get_counts(qc)
print(counts)
plot_histogram(counts)
plt.show()
```

- The circuit is one qubit plus one classical bit.
- Hadamard prepares superposition.
- Measurement is repeated 1000 times.
- `plot_histogram` visualizes the counts.

# Interpreting the Recorded Output

{1: 518, 0: 482}

- The observed distribution is close to uniform.
- About 51.8% of shots measured 1; 48.2% measured 0.
- The imbalance is ordinary sampling variation.

## **State preparation $\rightarrow$ measurement $\rightarrow$ repeated statistics**

- The qubit is not repeatedly “storing” both 0 and 1 as classical values.
- The quantum state is prepared once per shot and measured.
- The classical histogram summarizes many shots.

# Hands-on Activity

- Run the notebook several times.
- Compare the count imbalance from run to run.
- Increase shots and observe whether the histogram becomes more balanced.

## Lesson Summary

- Hadamard creates equal superposition from  $|0\rangle$ .
- Measurement is probabilistic.
- Histograms convert many measurement results into an intuitive visual.

**Master measurement statistics before moving to entanglement.**

## Quick Check

- ❶ What stays the same between runs?
- ❷ What changes?
- ❸ How does increasing shots change the histogram?